Ex.No.14 To Analyze the different types of servers using Webalizer tool

Aim

To install, configure, and execute the Webalizer log analysis tool on a Windows operating system to analyze raw server logs and generate interactive HTML dashboards containing traffic graphs and usage statistics.

Apparatus Required

1)Operating System: Windows 10 or Windows 11 (64-bit)

2)Software Bundle: XAMPP for Windows (includes Apache Web Server and pre-configured Webalizer)

3)Access Level: Windows Administrator privileges

4)Web Browser: Microsoft Edge, Google Chrome, or Mozilla Firefox

Theory

Server log files record every user transaction, but they are difficult to read in their raw .log text formats. Webalizer solves this by converting text logs into structured HTML pages. While Webalizer is natively a Linux tool, it is seamlessly integrated into Windows environments using XAMPP. XAMPP automatically pipes the Apache web server log (access.log) into the Webalizer engine. Webalizer then calculates key performance indicators:

1. Hits: Total number of files requested (images, text, pages).
2. Files: Total number of successful requests fulfilled by the server.
3. Pages: Pure HTML or PHP document views (ignoring individual embedded images).
4. Visits: Distinct user journeys tracked via unique IP addresses.

Step-by-Step Procedure

Step 1: Install the XAMPP Server Environment

1. Download the XAMPP installer for Windows from the official website.
2. Double-click the installer executable file.
3. During the Select Components prompt, ensure that Apache and Webalizer are checked.
4. Complete the wizard to install XAMPP into the default directory: C:\xampp\.

Step 2: Start the Web Server

1. Open the Windows Start Menu, search for XAMPP Control Panel, and run it as an Administrator.
2. Click the Start button next to the Apache module. The module background will turn green, indicating the web server is actively running and generating log files.

Step 3: Locate the Log and Configuration Files

Verify where Windows is storing your logs and how Webalizer is targeting them:

a) Raw Server Logs Location: C:\xampp\apache\logs\access.log

b)Webalizer Configuration File Location: C:\xampp\webalizer\webalizer.conf

(Optional Tuning): If you open webalizer.conf using Notepad, you will observe that XAMPP has pre-written the paths using forward-slash formatting compatible with the tool:

text

LogFile c:/xampp/apache/logs/access.log

OutputDir c:/xampp/htdocs/webalizer

Use code with caution.

Step 4: Execute Webalizer Manually via Windows Command Prompt

1. Press Win + R, type cmd, and hit Enter to launch the Command Prompt.
2. Change the directory to the XAMPP Webalizer folder:

cmd

cd C:\xampp\webalizer

Use code with caution.

3. Run the executable tool to process the active Windows logs:

cmd

webalizer.exe
Use code with caution.
The command prompt will flash an information line confirming how many lines of text data it successfully read and converted.
Step 5: View and Analyze the HTML Reports
1. Launch your web browser.
2. Type the following address into the URL bar and press enter:
http://localhost/webalizer/
3. Click on the current calendar month link to view the visual traffic analysis charts, total system bandwidth usage, and data tables tracking user visits.
Step 6: Automate the Log Analysis using Windows Task Scheduler
To avoid running the command prompt manually every day, set up a background Windows **Task:**
1. Press the Windows key, search for Task Scheduler, and open it.
2. In the right-hand Actions menu, select Create Basic Task.
3. Name your task XAMPP Webalizer Update and select Daily as the frequency trigger.
4. Set the Action to Start a program.
5. Click Browse and target the executable path: C:\xampp\webalizer\webalizer.exe
6. In the Start in (optional) textbox, type C:\xampp\webalizer to ensure the program finds its default configuration parameters. Click Finish.

Conclusion

The Webalizer analysis program was successfully deployed and executed on the Windows OS environment via XAMPP. Raw, unorganized data entries in access.log were converted into comprehensive, easy-to-read web graphs. The setup provides administrators with a highly visual framework to evaluate server stress, incoming traffic spikes, and user distribution patterns.